

PREPARE-Extraction-Tool User Manual



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1. Introduction

Extracting and interpreting clinical concepts from unstructured medical records is time-consuming and resource-intensive, often diverting healthcare professionals from direct patient care and limiting the secondary use of clinical data. To address this, we developed a semi-automated end-to-end pipeline that improves efficiency and interoperability while minimizing manual effort for clinical staff. The pipeline consists of three components: semi-automated extraction of clinically relevant information from medical records, semantic clustering of extracted entities to consolidate concept variants and reduce redundancy, and mapping of clustered concepts to standardized OHDSI clinical vocabularies available at OHDSI Athena.

2. Prerequisites

Before you can install and run the PREPARE-Extraction-Tool, you need to install a few required tools/packages.

2.1 Docker Desktop

What is Docker?

Docker can be thought of as a self-contained box that holds everything the tool needs to run (code, libraries, settings...). When you start Docker, it opens this box and runs the tool inside it and it works separately from your computer, so the tool works regardless of your operating system, and it doesn't interfere with any other software you have installed.

How to install Docker Desktop?

Download and install Docker Desktop from the official Docker website:

Windows: <https://docs.docker.com/desktop/install/windows-install/>

macOS: <https://docs.docker.com/desktop/install/mac-install/>

Linux: <https://docs.docker.com/desktop/install/linux-install/>

Once installed, open the Docker Desktop and leave it running in the background. You will know it's running when you see a small Docker icon appear in your system tray (Windows) or menu bar (Mac). As long as the icon is visible, Docker is ready.

Note: Docker Desktop requires a 64-bit operating system. On Windows, you may be prompted to enable WSL 2 (Windows Subsystem for Linux) during installation → follow the on-screen instructions if so. The installer will guide you through it.

2.2 The GLiNER model files (model.zip)

The tool uses GLiNER models for term extraction from medical text. The model is not included in the repository and must be obtained separately.

Where to get it?

The model can be found in the shared files we sent over Dropbox. Check your email and the shared Dropbox links to obtain the model.zip files.

Once you have found the model.zip file

1. Download the .zip from Dropbox
2. Unzip

NOTE: *For Mac users who will download the .zip from Safari, the files will be unzipped automatically and saved as a folder, so the unzip step is not required*

3. Place the unzipped model folder inside the bioner folder of the repository. In the end, the model path should look like this: **PREPARE-Extraction-Tool/bioner/model/**

For general public

If you do not have access to the shared Dropbox files, the default model is publicly available on Hugging Face:

E3-JSI/gliner-multi-med-ner-synthetic-v1

(<https://huggingface.co/E3-JSI/gliner-multi-med-ner-synthetic-v1>)

This is the same base model used by the tool and is suitable for general medical NER across multiple languages.

To download and set it up:

1. Install the Hugging Face Hub CLI if you do not already have it:

```
pip install huggingface_hub
```

2. Download the model using the CLI:

```
huggingface-cli download E3-JSI/gliner-multi-med-ner-synthetic-v1  
--local-dir bioner/model
```

Run this command from the root of the repository (**PREPARE-Extraction-Tool/**). The files will be saved directly into **bioner/model/**.

Alternatively, you can download the files manually from the Hugging Face model page by clicking Files and versions and downloading all files, then placing them in **bioner/model/**.

3. Verify the path. When done, the directory should look like this:

```
PREPARE-Extraction-Tool/bioner/model/
```

Once the model is in place, proceed with the Docker setup as described in section 2.3.

2.3 Vocabulary seed files

The tool maps extracted medical terms against the standard OHDSI vocabularies. Loading these vocabularies is optional during first-time setup, as you can also upload them later through the application's own interface, but having them ready in advance will save you time.

Where to get them?

Like the model files, we shared the vocabulary seed data via Dropbox. You can find it in the vocabulary folder, as `seed_data.zip` and `big_seed_data.zip`.

`seed_data.zip` contains a standard selection of OHDSI vocabularies sufficient for most use cases. `big_seed_data.zip` contains the full set of vocabularies available from OHDSI Athena and requires significantly more disk space. If you are unsure which one to use, start with `seed_data.zip` and upgrade to `big_seed_data.zip` if needed.

Unzip and place into the repository folder

Once you have located the vocabulary .zip files, do the same unzipping process as for the models.

In the unzipped folder, you should see 3 items:

- `vocabulary.csv` — the OHDSI vocabulary reference file
- `concept.csv` — the OHDSI concept definitions
- `es_repo` — a folder containing the Elasticsearch index data

Once you have these files, place them inside a folder named `seed_data` in the root of the repository, so the structure looks like:

```
PREPARE-Extraction-Tool/seed_data/vocabulary.csv  
PREPARE-Extraction-Tool/seed_data/concept.csv  
PREPARE-Extraction-Tool/seed_data/es_repo/
```

3. Installation & First-Time Setup

This section walks you through setting up the PREPARE-Extraction-Tool for the first time. Follow the steps in order. You only need to do this once. After doing all the steps, starting the tool takes a single command.

3.1 Get the repository files onto your computer

The tool's code lives in a Git repository on GitHub: <https://github.com/E3-JSI/PREPARE-Extraction-Tool>. You have two options, depending on whether you use Git.

Either way, the result is the same: a folder on your computer containing all the project files.

Option A: Download as a ZIP

Go to the repository page at <https://github.com/E3-JSI/PREPARE-Extraction-Tool>, click the green Code button, and select Download ZIP. Once downloaded, extract the ZIP to a location of your choice, for example, your Documents folder. You should end up with a folder named **PREPARE-Extraction-Tool-main**. Rename it to **PREPARE-Extraction-Tool** for simplicity.

Option B: Clone with Git

If you have Git installed, open a terminal, navigate to the folder where you want to place the project, and run:

```
git clone https://github.com/E3-JSI/PREPARE-Extraction-Tool
cd PREPARE-Extraction-Tool
```

3.2 Configure environment variables

The tool uses a configuration file called `.env` to know where its different parts are running. A ready-made template is already included in the repository.

To set up the environment open terminal inside **PREPARE-Extraction-Tool** folder and run:

```
cp .env.example .env
```

NOTE: if you are using Windows (Command Prompt) use:

```
copy .env.example .env
```

3.3 Place the GLiNER model files

If you have not done this already (covered in Section 2.2), place the extracted `model` folder inside the **bioner** directory of the repository:

```
PREPARE-Extraction-Tool/
├── bioner/
│   └── model/      ← your extracted model folder goes here
```

The tool will not start correctly without this folder in place.

3.4 Start all services

Make sure Docker Desktop is open and running (you should see its icon in your taskbar or menu bar). Then, in a terminal inside the **PREPARE-Extraction-Tool** folder, run:

```
docker compose up -d
```

This command downloads and builds all the necessary components (database, backend, frontend, and extraction service) and starts them in the background. The first time you run this, it may take up to 30 minutes, depending on your internet connection, because Docker needs to download the base packages. Subsequent starts are much faster.

NOTE: If `docker compose` error *The term 'docker compose' is not recognized as the name of a cmdlet, function, script file, or operable program. Check the spelling of the name, or if a path was included, verify that the path is correct and try again.* arises, just close the terminal and Docker; restart the computer and run it again.

When the command finishes and you see output confirming the containers are running, proceed to the next step.

3.5 Apply database migrations

This step sets up the database structure that the tool uses to store your work. Run:

```
docker compose exec backend alembic upgrade head
```

You only need to run this once after the first `docker compose up`. It is safe to run again later if you update the tool to a newer version.

3.6 (Optional) Load medical vocabularies - recommended

The tool maps extracted terms against standard OHDSI medical vocabularies. You can load these now via a script, or skip this step and upload them manually through the application interface later.

To load now via script (requires the seed files described in Section 2.3 to be in the **seed_data** folder):

```
./scripts/seed.sh
```

If you get a `permission denied` error, make the script executable first with

```
chmod +x scripts/seed.sh
```

then run it again.

On Windows, run this from Git Bash or WSL rather than PowerShell or Command Prompt. The script needs a Bash shell and direct access to the Docker socket on the host, so it cannot be run from inside the backend container.

This process populates the database and search index with vocabulary and concept data. It may take a while, depending on the size of the vocabulary files.

To skip and load later: *leave this step for now. Section 6 explains how to upload vocabularies through the application interface once the tool is running.*

3.7 Verify the installation

Open a web browser and go to:

<http://localhost:3000>

You should see the PREPARE Extraction Tool interface. If the page loads, your installation is complete and working.

If the page at `http://localhost:3000` does not load, check that Docker Desktop is still running and that the `docker compose up -d` command completed without errors. If there are no errors but the tool still doesn't run check Section 7 (Troubleshooting) which covers the most common issues.

4. Starting the Tool After First Setup

Once the initial setup from Section 3 is complete, starting the tool on subsequent uses is straightforward. You do not need to repeat any of the installation steps.

4.1 Start Docker Desktop

Before anything else, make sure Docker Desktop is open. Look for the Docker icon in your system tray (Windows) or menu bar (Mac). If it is not there, open Docker Desktop from your applications and wait until the icon appears and shows as running → it may take up to a minute to start.

4.2 Start the tool

Once Docker Desktop is running, open a terminal inside the **PREPARE-Extraction-Tool** folder and run:

```
docker compose up -d
```

Wait for the command to finish — this usually takes under a minute since everything is already downloaded. Then open your browser and go to `http://localhost:3000`.

One thing to keep in mind: even after the command finishes, the tool may not be fully ready yet. The frontend usually boots quickly, so the page at `http://localhost:3000` might load right away, but some of the other containers can still be starting up in the background. If you cannot log in at first, just wait a minute or two and try again.

4.3 Stop the tool when you are done

Always stop the tool before you close the terminal, quit Docker Desktop, or shut down your computer. Open a terminal inside the **PREPARE-Extraction-Tool** folder and run:

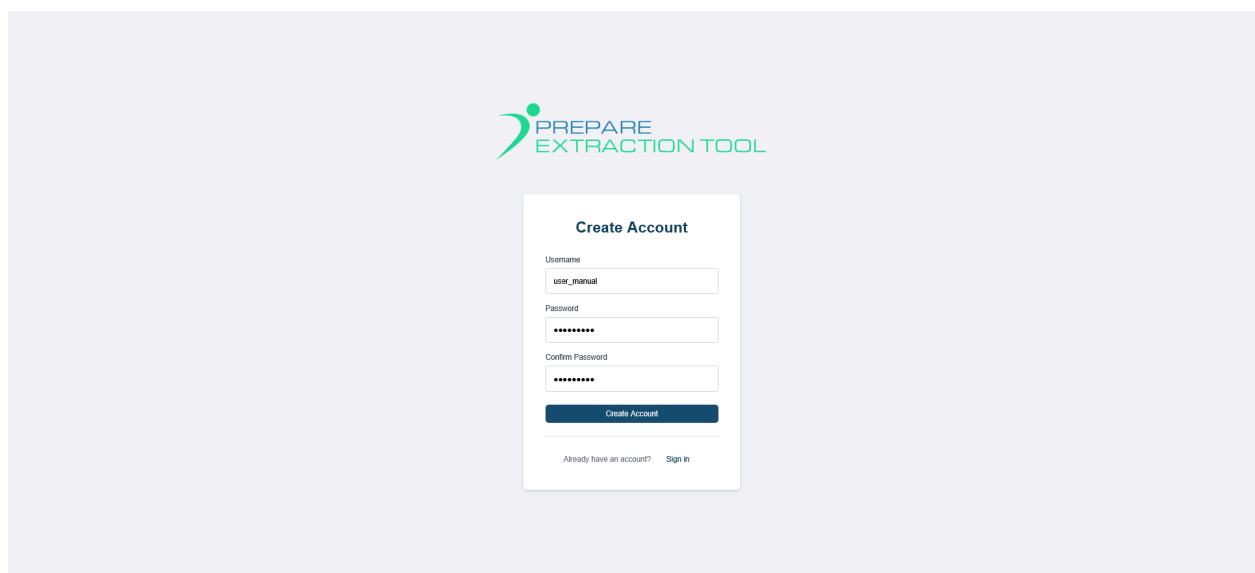
```
docker compose down
```

This stops all services cleanly without deleting any of your data. Your mappings, uploaded files, and vocabulary data are all saved in the database and will be there the next time you start the tool.

In short: open Docker Desktop and run `docker compose up -d` each time you start, and run `docker compose down` each time you finish.

5. Navigating the Interface (with edited screenshots)

5.0. Login page



The first page you will see when opening the tool is the login page. Before you can sign in for the first time, you need to create an account. Click **Create Account** and enter a username and password of your choice. Once your account has been created, use those same credentials to sign in every time you use the tool.

5. 1. Dataset upload

To begin, navigate to the **Datasets** page and click **+ Upload dataset**. You will be taken to the Upload Dataset form.

Here fill in the following fields:

Dataset name: Enter a unique name to identify your dataset (e.g. univpm_USAGI_test) .

Labels: Type each annotation category you want to use and press **Enter** or , to add it. Labels represent the types of medical entities that will be identified in the text, such as diagnosis, symptom, drug, date, or visit. You can also paste comma-separated values. Remove a label by clicking the × next to it.

Label relationships (optional): If some labels are semantically related to each other, check the box labelled *Some labels are related to each other*. This reveals the Label relationships section, where you can define which label has a value of another label (ex. *Test has value Punteggi Test*). Click **+ Add relationship** to add more pairs. When a relationship is defined, you are able to link individual annotations across those two labels in the annotation panel using the **Link** button (look at the **5.5. Annotation panel section**).

This feature is useful when a clinical concept is only meaningful together with its associated value (ex. a test name such as *NMSS* paired with its score *80/360*). Defining a relationship between a *Test* and a *Punteggi Test* label allows you to explicitly link the two, preserving the concept-value pair in the extracted data. This mirrors the structure of OMOP CDM domains such as Measurement and Observation, where a concept is always recorded alongside its value, making linked annotations easier to map correctly in the Concept Mapping step.

Date label (optional): Select which of your labels corresponds to dates. Extracted terms will inherit the timestamp from the nearest annotation with this label. If left blank, the tool falls back to the record's `visit_date` column or the upload time.

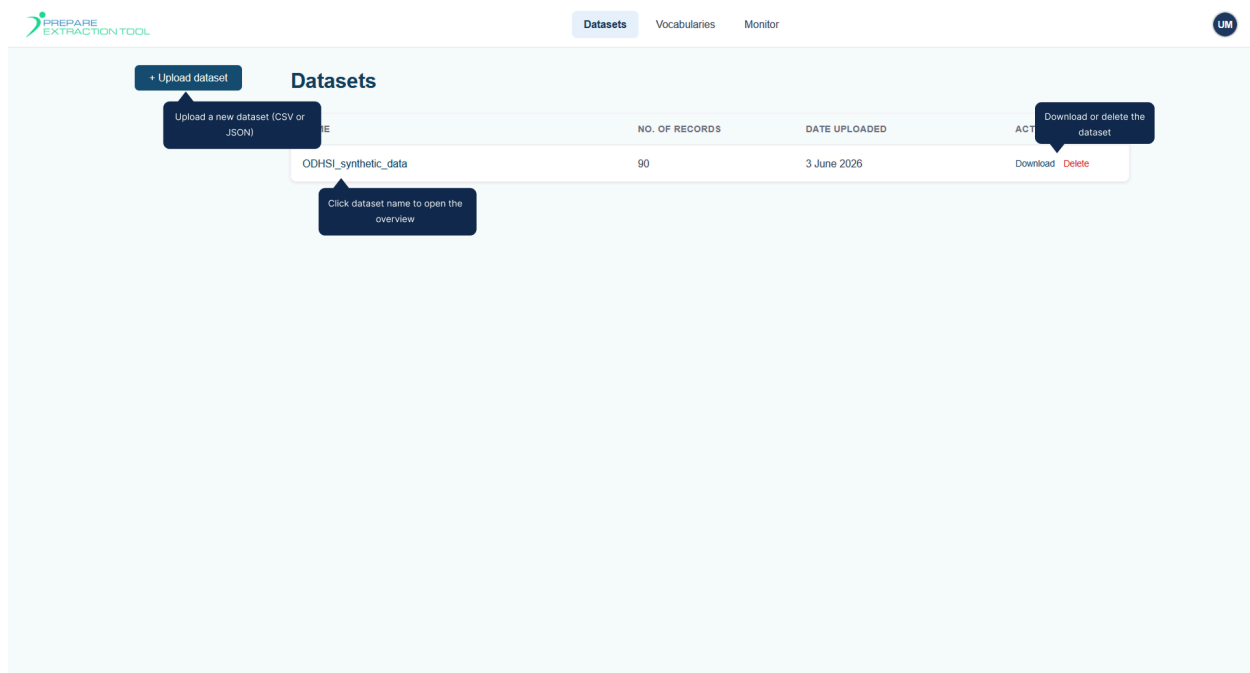
Upload dataset file: Drag and drop your file onto the upload area or click **Browse for file**. The file must be in CSV or JSON format and must include the following columns:

- **patient_id** → the patient identifier
- **visit_date** → the date of the visit
- **text** → the full text of the clinical record

An optional `seq_number` column can be included if a long record has been split into multiple observations, to identify the sequence of each part. Maximum file size is 2 GB.

Once all fields are filled in, scroll down and click **Upload Dataset** to submit.

5. 2. Dataset list



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The home page displays all uploaded datasets. For each dataset, the number of records and the upload date are shown.

Use the **+ Upload dataset** button to begin uploading a new dataset, and the **Download** and **Delete** buttons to export or remove an existing one. Clicking on a dataset name redirects you to the dataset overview with workflow steps.

5. 3. Dataset overview

The screenshot shows the 'Dataset Overview and Statistics' page for a dataset named 'ODHSI_synthetic_data'. At the top, there are navigation tabs: 'Datasets' (selected), 'Vocabularies', and 'Monitor'. Below the dataset name, there are labels: 'diagnosis', 'date', 'symptom', 'test score', 'visit', 'Comorbidity', and 'drug'. The upload date is '3 June 2026'. A summary bar shows: 90 Total Records, 0 Records Processed, 0 Total Terms, and 90 Pending Review. Below this, three workflow steps are displayed as cards:

- 1st step: Term Extraction** (Extract medical entities from clinical text):
 - Total Records: 90
 - Terms Extracted: 0
 - PROGRESS: 0 / 90 (0%)
 - Buttons: View Records, Extract All
- 2nd step: Term Clustering** (Group similar terms for standardization):
 - Clusters Created: 0
 - Clustered Terms: 0
 - Unclassified Terms: 0
 - Buttons: View Clusters, Auto-Cluster
- 3rd step: Concept Mapping** (Map clusters to standard vocabulary concepts):
 - Total Clusters: 0
 - Mapped Clusters: 0
 - Unmapped Clusters: 0
 - PROGRESS: 0 / 0 (0%)
 - Buttons: View Mappings, Start Mapping

At the bottom, there is a copyright notice: '© 2023-2027 PREPARE Project.'

After uploading, go to the **Dataset Overview** page by clicking on the dataset name in the **Dataset List**.

This section serves as the central hub for your dataset. At the top you can see the dataset name, upload date, and all configured labels. The summary bar shows the total number of records, how many have been processed, the total number of extracted terms, and how many records are still pending review.

Below the summary, the three workflow steps are displayed as cards:

- **1st step Term Extraction:** Extract all labels with the NER model on all records. Click **Extract All** to start automated extraction, or **View Records** to work through records manually.
- **2nd step Term Clustering:** Auto-cluster all extracted terms into groups of similar terms. Click **Auto-Cluster** to run grouping automatically, or **View Clusters** to manage clusters manually.

- **3rd step Concept Mapping:** Map the clusters to OMOP CDM vocabularies. Click **Start Mapping** to begin automated or manual concept mapping, or **View Mappings** to review existing mappings.

Each card displays its own progress statistics so you can track completion across all three steps at once.

5. 4. Term Extraction

Go to **Term Extraction** either from the Dataset Overview (**View Records**) or by clicking **Extract All** to trigger automated processing in the background.

The page is divided into three panels:

Left panel → Records List. All patient records are listed here. Click any record to open it in the central NER view. Records that have already been reviewed are marked with a **Reviewed** badge. You can filter the list to show all records, reviewed records only, or unreviewed records only. You can also search for the records based on their patient ID or title.

Central panel → NER View. The selected record is displayed with all identified annotations highlighted in colour. Each colour corresponds to one annotation label (e.g. red for diagnosis). You can monitor the overall extraction progress in the top-right status bar, which shows how many records have been processed (e.g. *27/90 records · 30%*). Click **Cancel** if you need to stop the extraction process.

Right panel → Extracted Terms. All terms identified in the currently selected record are listed here with their assigned label, timestamp and linked entities.

To manually review or edit the annotations in a record, click **Edit Labels** to open the **Annotation Panel**.

5. 5. Annotation panel

The screenshot displays the PREPARE EXTRACTION TOOL interface, specifically the Annotation Panel and Records List. The Records List on the left shows a list of patient records with search filters and a 'REVIEWED' status. The Annotation Panel on the right is divided into two main sections: 'Labels' and 'Annotations (15)'. The 'Labels' section includes a 'Link' button and a list of labels (Sintomi, Farmaco, Dose del farmaco, Events, Comorbidità, Test, Data, Punteggi Test, Riabilitazione). The 'Annotations (15)' section shows a list of annotations, including 'Malattia di Parkinson idiopatica', 'dal 2009', '18/03/2021', and 'Attiva PC'. The 'Annotations (15)' section also includes a 'Mark as Reviewed' button and a 'Delete' button. The 'Annotations (15)' section is further divided into 'Creating annotations' and 'Changing labels' sections. The 'Creating annotations' section includes instructions on how to create annotations and a list of keyboard shortcuts. The 'Changing labels' section includes instructions on how to change labels and a list of keyboard shortcuts. The 'Annotations (15)' section also includes a 'Delete' button and a 'Mark as Reviewed' button. The 'Annotations (15)' section is further divided into 'Creating annotations' and 'Changing labels' sections. The 'Creating annotations' section includes instructions on how to create annotations and a list of keyboard shortcuts. The 'Changing labels' section includes instructions on how to change labels and a list of keyboard shortcuts.

The Annotation Panel opens as an overlay when you click **Edit Labels** in the NER view. It allows you to add, edit, delete, and link annotations within individual clinical records.

Adding an annotation:

Select a label from the label bar at the top of the panel (labels are numbered 1–9 and colour-coded). Then drag over the target word or phrase in the record text to highlight it — an annotation is created instantly. Alternatively, press the number key corresponding to your desired label before dragging.

Changing a label:

Click any highlighted annotation in the text to select it, then click a different label in the label bar or press the corresponding number key (1–9). The label updates immediately.

Deleting an annotation:

Click an annotation to select it, then press Delete or Backspace. This also removes all links associated with that annotation.

Entity linking:

If label relationships have been configured for the dataset, you can link related annotations to each other. Click on an annotation to select it, then click Link — this opens linking mode. In linking mode, click on any compatible term in the text (shown in green) to link it to the selected annotation. Orange highlights indicate terms that are already linked. To unlink, click the orange term. Press Esc to exit linking mode without saving. In the right sidebar, linked annotations display the relationship (e.g. linked to 18/3/2021).

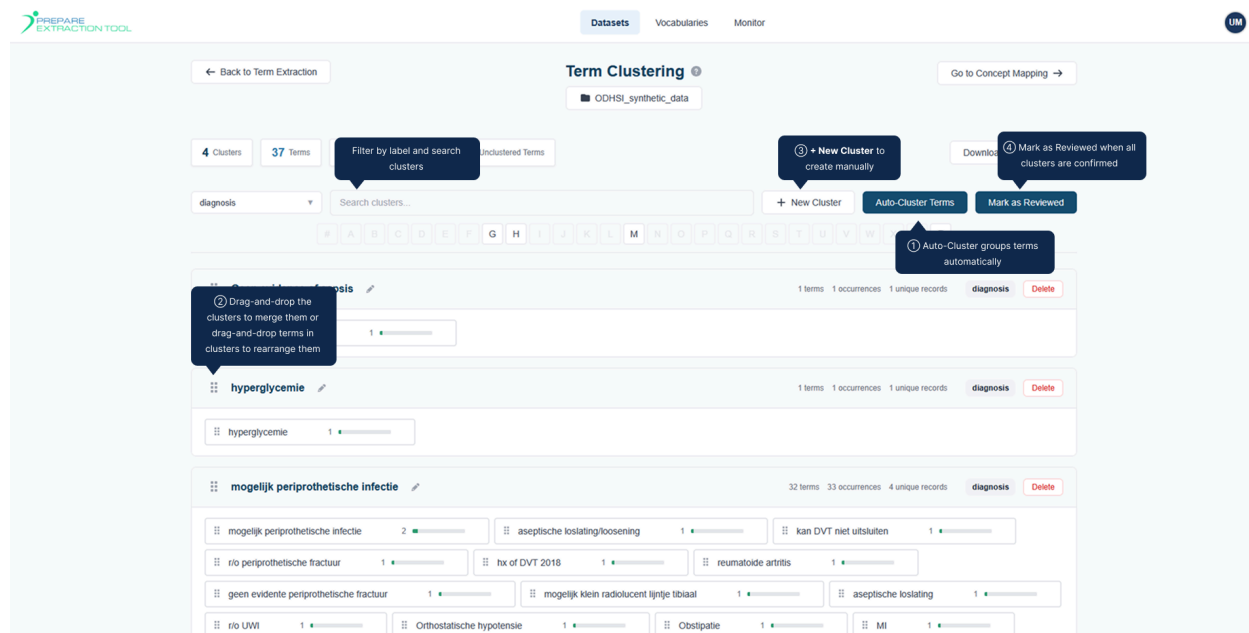
Navigating between records:

Use the Previous and Next buttons in the top-right corner, or press the ← and → keyboard shortcuts, to move between records without closing the panel.

Marking a record as reviewed:

Once you are satisfied with all annotations in the current record, click Mark as Reviewed or press Enter. The button turns green to confirm, and the tool advances to the next record automatically.

5. 6. Term Clustering



Once extraction is complete or sufficiently advanced, navigate to Term Clustering via **Go to Term Clustering** or from the Dataset Overview.

When you first arrive on this page, all extracted terms are unclustered so they appear in a single pool of ungrouped terms at the bottom of the page. No clusters exist yet until you either run **Auto-Cluster** or **create them manually**. The goal of this step is to move all terms out of this unclustered pool and into meaningful groups before proceeding.

The clustering page works one label at a time. Use the label dropdown in the toolbar to select which entity category you are currently working on (e.g. Diagnosis, Drug, Symptom). All statistics, clusters, and actions on the page apply only to the selected label. The summary bar shows the number of clusters, total terms, and unclustered terms for the currently selected label. Aim for zero unclustered terms in each label before moving on.

1. **Auto-clustering:** Click **Auto-Cluster Terms** to automatically group semantically similar terms into clusters. This is the fastest way to organise a large number of extracted terms and the recommended starting point before any manual adjustments.
2. **Reviewing and editing clusters:** Each cluster shows its name, the number of terms it contains, total occurrences, unique records, and its label category. To manually rearrange content, drag and drop individual terms between clusters to move them, or drag entire cluster headers to merge two clusters together.
3. **Renaming a cluster:** Click the pencil icon next to the cluster name to rename it.
4. **Creating a cluster manually:** Click **+ New Cluster** to create an empty cluster, then drag terms into it from other clusters.

5. **Filtering and searching:** Use the label dropdown on the left to filter clusters by annotation category. Use the search bar to find a specific cluster by name. You can also use the alphabetical index to jump to clusters starting with a particular letter.
6. **Deleting a cluster:** Click **Delete** on the right side of a cluster header to remove it. Its terms will return to the unclustered pool.
7. **Downloading cluster data:** Click **Download Term Clusters** to export the current cluster state as a file.

Work through every label in the dropdown in turn: select a label, run Auto-Cluster and adjust manually, then click **Mark as Reviewed**. The label locks to read-only and the button turns green. Switch to the next label and repeat. Only when all labels have been marked as reviewed should you proceed to Concept Mapping using the **Go to Concept Mapping** button in the top right.

5. 7. Concept Mapping

The screenshot displays the 'Concept Mapping' interface of the PREPARE EXTRACTOR TOOL. At the top, there's a navigation bar with 'Datasets', 'Vocabularies', and 'Monitor' tabs, and a 'UM' user icon. Below this, a 'Back to Term Clustering' button is on the left, and a 'Back to Dataset Overview' button is on the right. The main header shows 'Concept Mapping' with a dropdown for 'ODHSI_synthetic_data'. A summary bar indicates '4 Total', '0 (0%) Mapped', '0 Approved', and '4 Unmapped'. There are buttons for 'Auto-Map All Terms' and 'Download Mappings'. A dropdown menu is set to 'diagnosis'. The main table lists source terms with columns: STATUS, SOURCE TERM, FREQ, LABEL, CONCEPT ID, CONCEPT NAME, DOMAIN, VOCABULARY, and ACTIONS. Callouts provide instructions: 1. 'Auto-Map runs automated concept matching' points to the 'Auto-Map All Terms' button. 2. 'Accept or Remove each suggested mapping' points to the 'Accept' and 'Remove' buttons in the ACTIONS column. 3. 'Click a row to load target concept suggestions below' points to a row in the table. 4. 'Panel shows target concepts for the selected source term' points to the 'Target Concepts' section at the bottom. The 'Target Concepts' section includes a 'QUERY MODE' section with checkboxes for 'Standard concepts only' and 'Include source terms', and a 'FILTERS' section with dropdowns for 'VOCABULARY', 'DOMAIN', and 'CONCEPT CLASS'. The 'Target Concepts' table has columns: SCORE, CONCEPT ID, CONCEPT NAME, DOMAIN, CONCEPT CLASS, VOCABULARY, CONCEPT CODE, and STANDARD.

The Concept Mapping page links each cluster to a standardised concept in the OMOP CDM vocabulary. The summary bar shows the total number of clusters, how many have been mapped and approved, and how many remain unmapped.

Automated mapping: Click **Auto-Map All Terms** to run automated concept matching across all clusters. The tool searches the Elasticsearch index and assigns the highest-scoring candidate concept to each cluster. *Note that this button requires vocabularies to be loaded (see section 6).*

Manual mapping: Click on any row in the cluster table to select it. The bottom panel labelled **Target Concepts** will populate with a ranked list of candidate concepts retrieved from the vocabulary, sorted by similarity score. Each candidate shows its Concept ID, Concept Name, Domain, Concept Class, Vocabulary, Concept Code, and whether it is a standard concept.

Review the suggested concepts and click **Accept** next to the correct match to confirm the mapping, or **Remove** to reject it. You can add a free-text comment to any mapping using the **Comment** field above the bottom panel.

Filtering and searching: Use the label dropdown to filter the cluster table by annotation category (e.g. show only diagnosis clusters). Use the search bar to find a specific source term.

Query mode and filters: In the bottom panel, you can switch between **Use source term** (the default, which searches using the cluster name) and **Custom query** (which lets you type an alternative search term). You can also apply filters to restrict results to standard concepts only, include source terms in the search, and filter by Vocabulary, Domain, or Concept Class.

Downloading mappings: Once all clusters have been reviewed and approved, click **Download Mappings** to export the results.

5.8 Monitoring Dashboard

The Monitoring Dashboard lets you train NER models based on your annotated datasets to improve overall entity extraction performance. It is organised into two tabs: **Training**, where new fine-tuning runs are configured and launched, and **Models**, where completed runs are inspected and evaluated.

The screenshot displays the 'Monitoring Dashboard' interface, specifically the 'Training' tab. The dashboard is organized into several sections:

- Training data:** Includes a 'Select datasets...' dropdown and an 'Evaluation datasets (optional)' section with a 'Held-out split of training data' dropdown.
- Base model:** Features a 'Multilingual (default): 299M ~8 GB VRAM' dropdown and a table listing available models.
- Train / Eval split:** Includes a '90 / 10' dropdown and an 'Advanced (hyperparameters)' section with 'Start' and 'Stop' buttons.
- Training progress:** A progress bar at the bottom showing 0% completion.

Annotations with numbered circles are overlaid on the interface:

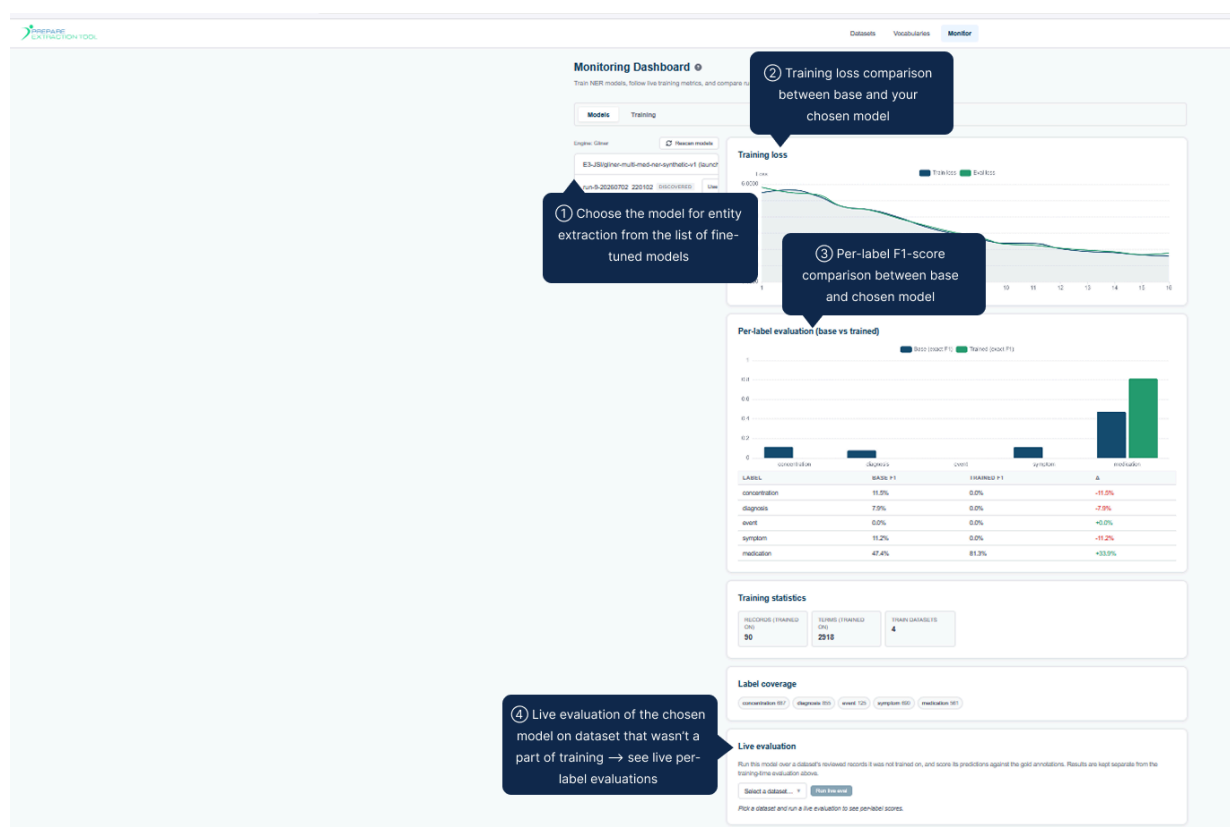
- ① Chose the training dataset (this is one of your reviewed datasets)
- ② Chose the model you want to fine-tune. It can be one of our models or any other GINER compatible model
- ③ Chose the training/evaluation split
- ④ Start the training
- ⑤ Follow live training progress

Configuring a training run: Under the **Training** tab, select a **Training dataset** from the dropdown menu; meaning one of your reviewed datasets. Optionally, choose an **Evaluation dataset** to score the model against a held-out set during training; if no dataset is selected for evaluation, a held-out split of the training data is used instead.

Choosing a base model: In the **Base model** section, select the model you want to fine-tune. This can be one of the tool's built-in models or any other GLiNER-compatible model.

Train/Eval split and hyperparameters: Choose the **Train/Eval split ratio** (e.g. 90/10) to control how much of the training data is held back for evaluation. Expand **Advanced** (hyperparameters) if you need to adjust training settings such as num_epochs, training_speed and others.

Starting training: Click **Start** to launch the training run, or **Stop** to cancel a run in progress. Once started, the **Training progress** bar below shows live completion percentage as the run proceeds. After the training is complete, the model will be added to the models list in the **Models** tab.



Reviewing trained models: Switch to the **Models** tab and select a fine-tuned model from the list to review its performance. Each entry shows the model name and run identifier.

Training loss: The Training loss chart compares the base model's loss against your chosen model's loss over training steps, letting you assess whether fine-tuning improved.

Per-label evaluation: The **Per-label evaluation** (base vs trained) chart and accompanying table break down exact-match F1 scores by entity label, comparing the base model's score against the trained model's score for each label. Use this to identify which labels benefited most from fine-tuning and which may have regressed.

Training statistics and label coverage: The **Training statistics** panel summarises the number of records trained on, terms trained on, and the number of training datasets used. The Label coverage panel shows how many annotated instances of each label were available during training.

Live evaluation: Use the Live evaluation panel to run the chosen model over a dataset's reviewed records that it was not trained on, scoring its predictions against the gold annotations. Select a dataset from the dropdown and click Run live eval; results are kept separate from the training-time evaluation shown above and display per-label scores once complete.

6. Loading Vocabularies Through the UI

For users who skipped the `seed.sh` script during initial setup.

The Concept Mapping step requires medical vocabulary data to be loaded into the tool before automated or assisted mapping can function. If this was not done during setup, navigate to the **Vocabularies** page using the link in the top navigation bar.

You will need three items obtained from the shared setup package described in the repository:

- **vocabulary.csv**: metadata for all vocabulary sources (SNOMED CT, RxNorm, LOINC, ICD10, etc.)
- **concept.csv**: all OMOP standard concepts with their IDs, names, domains, and concept classes
- **es_repo/**: a pre-built Elasticsearch snapshot that enables full-text concept search and scoring

Upload these in order: `vocabulary.csv` first, then `concept.csv`, then the `es_repo/` folder. Wait for each upload to complete and confirm a successful status before proceeding to the next. The `concept.csv` file is large and may take several minutes to process.

Once all three show a successful status, return to your dataset and proceed with Concept Mapping. The Vocabularies page does not need to be revisited unless you wish to update to a newer version of the OHDSI vocabularies.

To obtain **vocabulary.csv** and **concept.csv** and **es_repo/** folder use the Dropbox link we sent to you via E-mail.

7. Troubleshooting

This section covers the most common issues encountered when setting up and running the PREPARE Extraction Tool, along with steps to resolve them.

7.1 Model not loading

Symptom: Term Extraction fails to start, the extraction service crashes on startup, or NER annotations are not produced when running extraction on records.

Verify the model files are in the correct location. The GLiNER model must be placed inside the `bioner/` directory before the containers are started. The expected path is:

```
bioner/model/
```

The `model/` folder should be extracted from the shared `model.zip` file referenced in the repository. If you are using a fine-tuned model, place that extracted folder in the same location, replacing the default one.

Common mistakes to check:

- The folder is named something other than `model` (e.g. `model-finetuned/` or a nested subfolder like `model/model/`)
- The zip was extracted but the resulting folder was placed in the project root instead of inside `bioner/`
- The `bioner/` container was built before the model files were added - in this case you need to rebuild:

```
docker compose down
docker compose up -d --build
```

7.2 Finding logs

Logs are the fastest way to diagnose any issue not covered by the steps above. Use the following commands to view output from individual services:

```
# View logs for a specific service
docker compose logs -f backend
docker compose logs -f frontend
docker compose logs -f bioner
```

7.3 Quick fixes

Before investigating further, try the following in order - they resolve the majority of transient issues:

Refresh the page. Some UI states (stuck progress bars, blank panels, unresponsive buttons) are caused by a stale browser session. A hard refresh (**Ctrl+Shift+R** on Windows/Linux, **Cmd+Shift+R** on macOS) clears cached state.

Log out and log back in. If the interface is behaving unexpectedly after a period of inactivity, your session token may have expired. Click your profile icon in the top right and log out, then log back in.

Restart the Docker services. If the application becomes unresponsive or a service is behaving inconsistently, a clean restart often resolves it:

```
docker compose down
docker compose up -d
```

Check available disk space. If uploads fail silently or the database stops accepting writes, the host machine may have run out of disk space. The vocabulary files and Docker images together can require several gigabytes. Run **df -h** (macOS/Linux) or check **Disk Management** (Windows) to verify.

8. References

- API documentation (<http://localhost:8000/docs>)
- Database admin panel (<http://localhost:8080>)
- Dropbox:
<https://www.dropbox.com/scl/fo/c2r6z0av5ezfixaye59a4/AJRcziY3oV4Qt1ahbcNLKaY?rlkey=arsq5ihbkgu4o8wuyemhfcwpw&e=1&st=c3of09gk&dl=0>
- GitHub repository: <https://github.com/E3-JSI/PREPARE-Extraction-Tool>